

# OPEN QUANTUM SAFE

## Community Health & Contribution Analysis

liboqs · oqs-provider

Analysis date: June 25, 2026 | Data through: June 25, 2026 | Period covered: 2016–2026

|                                                                                      |                                                                                                |                                                                                  |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <div>liboqs Health Score</div> <div>6.5 / 10</div> <div>Maturing · Stabilising</div> | <div>oqs-provider Health Score</div> <div>5.1 / 10</div> <div>Recovering · Transitioning</div> | <div>Combined Ecosystem</div> <div>5.9 / 10</div> <div>Structural Progress</div> |
|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|

Scores assess community breadth, retention, growth trend, and bus factor — not code quality.

## Executive Summary

**The OQS ecosystem has achieved its most significant governance improvements since the project’s founding.** Since the March 14, 2026 report, three critical governance recommendations have been fully implemented: RodriM11 elevated to Maintainer in oqs-provider (G8), xuganyu96 added as Maintainer in liboqs (G7), and all three inactive Committers (SWilson4, Martyrshot, praveksharma) moved to Emeritus status (G1–G3). These are structural changes with measurable impact: RodriM11 now drives oqs-provider as its most active contributor, and xuganyu96 is the second most active contributor in liboqs.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>liboqs — Strengths</b> <ul style="list-style-type: none"><li>437 all-time contributors across 10 years (excl. bots)</li><li>Active streak of 43 consecutive months ≥10 contributors (Jun 2022–Jun 2026, ongoing)</li><li>Leadership rotation: dstebila leads the 6-month window at 20.9%; xuganyu96 second at 15.4%; baentsch third at 11.4% — healthiest distribution in project history</li><li>Bus factor improves to 4 on the 6-month window (was 3 in March)</li><li>xuganyu96 formalised as Maintainer — closes the succession gap identified in March</li></ul> | <b>oqs-provider — Progress &amp; Remaining Concerns</b> <ul style="list-style-type: none"><li>RodriM11 now leads the 6-month window at 37.2% (195 events); baentsch at 25.4% — significant redistribution from March's 47.0%</li><li>Merge authority now rests with two active Maintainers, not one — structural risk substantially reduced</li><li>Event volume declined –46.1% YoY (2024→2025) — severe, though stabilising in late 2025</li><li>Review quality standards remain undefined project-wide: empty-approval rates range from 36% to 99% across reviewers, with no documented minimum expectation (G10)</li><li>Contributor breadth still thin: 8–14 contributors/month throughout 2025–2026</li></ul> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Five critical priorities:** (1) Document baentsch’s LoA end date — the LoA now exceeds its 12-month governance maximum; (2) Establish a shared definition of review quality across both repos (G10); (3) Formalise abhi-dev-engg as Committer in liboqs (116 events in 6 months, no formal role); (4) Write Architecture Decision Records for oqs-provider; (5) Set a numeric target to bring oqs-provider’s top-1 concentration below 35%. See section 5 for the full ranked list of fifteen recommendations.

# 1. Dataset & Methodology

This report is generated from live GitHub REST API data pulled on June 25, 2026 using a read-only personal access token (public\_repo scope), using the same event types as the March 2026 report: issue comments, pull request review inline comments, PR review verdicts (approve / request changes / commented), PR lifecycle events (opened, merged via individual PR fetch for merged\_by), discussion threads and replies, and commit comments. Commits themselves are not counted; the analysis focuses on the community engagement layer.

Conventions matching the March report: total event counts include all events (including bot-authored commit comments, as bots account for the majority of commit comment volume); contributor counts for monthly tables show all unique human actors including departed contributors (SWilson4, praveksharma, Martyrshot); health metrics (bus factor, concentration shares, retention) exclude both bots and departed contributors. Bots excluded from contributor analysis: dependabot[bot], github-actions[bot], codecov[bot], pre-commit-ci[bot], coveralls, mergify[bot], and all accounts ending in [bot].

| Repository                     | Period              | Total Events | Contributors (excl. bots) | Months |
|--------------------------------|---------------------|--------------|---------------------------|--------|
| open-quantum-safe/liboqs       | Aug 2016 – Jun 2026 | 18,214       | 437                       | 119    |
| open-quantum-safe/oqs-provider | Feb 2021 – Jun 2026 | 5,658        | 217                       | 65     |
| Combined ecosystem             | Aug 2016 – Jun 2026 | 23,872       | 571*                      | 119    |

*\* 83 contributors are active in both repositories (any events); 571 is the deduplicated total. Total event counts include bot-authored commit comments; contributor counts exclude bots.*

## 2. liboqs — Detailed Analysis

### 2.1 Key Health Metrics

All metrics use a 4-year window (Jun 2022 – Jun 2026), excluding bots and departed contributors (SWilson4, praveksharma, Martyrshot), for a current rather than historical view of project health.

| Metric                      | Value             | Context                                                           |
|-----------------------------|-------------------|-------------------------------------------------------------------|
| Total interactions          | 9,189             | 4-year window (2022+), excl. departed & bots                      |
| Unique contributors         | 334               | 4-year window; 95 (28.4%) appear only once                        |
| Bus factor (4-year)         | 3                 | 3 people = 50% of 4-year activity                                 |
| Bus factor (6-month)        | 4                 | Improved: 4 people = 50% of recent activity                       |
| Top contributor (4yr)       | baentsch<br>27.4% | dstebila 23.0% — gap narrowing significantly                      |
| Top contributor (6mo)       | dstebila 20.9%    | xuganyu96 15.4%, baentsch 11.4% — healthiest distribution ever    |
| Top-3 share (4yr)           | 53.7%             | Moderate concentration, stable                                    |
| One-time contributors (4yr) | 28.4% (95)        | Structural funnel challenge, unchanged                            |
| Retained ≥3 months (4yr)    | 24.0% (80)        | 2023 cohort best at 34%; 2024–2025 cohorts at 19–21%              |
| Post-NIST contributors      | 207               | vs 169 pre-NIST (4yr) — community broadened post-standardisation  |
| Active streak ≥10 contrib.  | 43 months         | Jun 2022 – Jun 2026 — ongoing and stronger than March’s 40 months |
| Event trend (YoY 2024→2025) | –28.3%            | Post-NIST normalisation; breadth remains solid                    |
| New contributors 2024       | 103               | Record intake year (4yr window, excl. departed)                   |
| New contributors 2025       | 83                | Stable; 2026 on pace for ≈56 (partial year)                       |

### 2.2 Growth Narrative — Four Eras

**Era 1 — Origins (2016–2019):** The project emerged with a committed but tiny founding team. Activity was sparse and dominated by 3–5 people. The 2018 peaks reflect early algorithmic debates as NIST’s post-quantum competition gathered momentum. Only 8 of the first 61 contributors (13%) were still active a year later.

**Era 2 — COVID Surge (2020–2021):** Remote-work patterns and heightened attention to cryptographic infrastructure drove a step-change. Monthly contributor counts rose from single digits to 10–15, and the first true community structure emerged. The 2021 cohort showed the best retention of any year (36% active ≥3 months).

**Era 3 — NIST Countdown (2022–2024):** The approach of NIST’s final PQC standards transformed the project. Contributor intake: 42 new in 2022, 78 in 2023, a record 103 in 2024. June 2025 (29 contributors, 510 events) and Jul–Aug 2025 (30 and 33 contributors) represent the highest breadth months in project history.

**Era 4 — Post-Standardisation (2025–present):** Following NIST FIPS publication, activity moderated. Crucially, the active streak extended to 43 months (still ongoing in Jun 2026) and May 2026 recorded 406 events. The leadership rotation to dstebila and xuganyu96 suggests genuine community maturation beyond dependence on a single driver.

2.3 Monthly Activity — Last 18 Months

| Month   | Contributors | Events | Activity bar |
|---------|--------------|--------|--------------|
| 2024-12 | 20           | 144    | <div></div>  |
| 2025-01 | 33           | 334    | <div></div>  |
| 2025-02 | 27           | 291    | <div></div>  |
| 2025-03 | 22           | 205    | <div></div>  |
| 2025-04 | 14           | 166    | <div></div>  |
| 2025-05 | 23           | 98     | <div></div>  |
| 2025-06 | 29           | 510    | <div></div>  |
| 2025-07 | 30           | 380    | <div></div>  |
| 2025-08 | 33           | 303    | <div></div>  |
| 2025-09 | 21           | 343    | <div></div>  |
| 2025-10 | 22           | 264    | <div></div>  |
| 2025-11 | 20           | 225    | <div></div>  |
| 2025-12 | 13           | 88     | <div></div>  |
| 2026-01 | 28           | 249    | <div></div>  |
| 2026-02 | 25           | 259    | <div></div>  |
| 2026-03 | 12           | 300    | <div></div>  |

| Month    | Contributors | Events | Activity bar |
|----------|--------------|--------|--------------|
| 2026-04  | 13           | 276    | <div></div>  |
| 2026-05  | 19           | 406    | <div></div>  |
| 2026-06* | 21           | 213    | <div></div>  |

Events include all engagement types including bot-authored commit comments, matching March report convention. Contributors show unique human actors. \* June 2026 is a partial month at time of data export.

2.4 Key Contributors (6-Month Window: Dec 2025 – Jun 2026)

| Contributor   | Events | Share | Active Months | Notes                                                  |
|---------------|--------|-------|---------------|--------------------------------------------------------|
| dstebila      | 250    | 20.9% | 6/7           | Maintainer — now leading contributor across both repos |
| xuganyu96     | 184    | 15.4% | 7/7           | New Maintainer; Release Manager oqs-provider           |
| baentsch      | 136    | 11.4% | 6/7           | Maintainer; LoA status requires clarification (G5)     |
| abhi-dev-engg | 116    | 9.7%  | 4/7           | No formal role — G6 candidate                          |
| bhess         | 106    | 8.8%  | 6/7           | Committer — consistent                                 |
| ashman-p      | 69     | 5.8%  | 6/7           | Committer; see review standards discussion (G10)       |
| RodriM11      | 31     | 2.6%  | 5/7           | Committer (liboqs) / Maintainer (provider)             |
| rben-dev      | 28     | 2.3%  | 4/7           | Rising contributor                                     |
| mkannwischer  | 26     | 2.2%  | 7/7           | Consistent long-tenure contributor                     |
| rtjk          | 25     | 2.1%  | 2/7           | Regular; cross-repo                                    |

Events exclude bots and departed contributors. Shares are of the 6-month human-only total (1,198 events).

2.5 Cohort Retention Analysis

For each intake year, the fraction of contributors active in subsequent years and for ≥3 months. This is the primary indicator of community ‘stickiness’.

| First Year | New Contributors | Active yr+1 | Active yr+2 | ≥3 Months Active |
|------------|------------------|-------------|-------------|------------------|
| 2016       | 8                | 7 (88%)     | 4 (50%)     | 7 (88%)          |

| First Year | New Contributors | Active yr+1 | Active yr+2 | ≥3 Months Active           |
|------------|------------------|-------------|-------------|----------------------------|
| 2017       | 21               | 4 (19%)     | 1 (5%)      | 4 (19%)                    |
| 2018       | 14               | 4 (29%)     | 3 (21%)     | 7 (50%)                    |
| 2019       | 18               | 4 (22%)     | 3 (17%)     | 6 (33%)                    |
| 2020       | 28               | 9 (32%)     | 4 (14%)     | 9 (32%)                    |
| 2021       | 28               | 7 (25%)     | 9 (32%)     | 10 (36%)                   |
| 2022       | 38               | 5 (13%)     | 2 (5%)      | 5 (13%) <a href="#">↱</a>  |
| 2023       | 70               | 22 (31%)    | 13 (19%)    | 24 (34%)                   |
| 2024       | 103              | 22 (21%)    | 9 (9%)      | 20 (19%) <a href="#">↱</a> |
| 2025       | 81               | 11 (14%)    | —           | 17 (21%) <a href="#">↱</a> |

The 2024–2025 cohorts at 19–21% three-month retention remain below the 30–36% achieved in 2019–2021, confirming that onboarding friction persists despite record intake volumes.

### 3. oqs-provider — Detailed Analysis

#### 3.1 Key Health Metrics

All metrics use a 4-year window (Jun 2022 – Jun 2026), excluding bots and departed contributors.

| Metric                      | Value                 | Context                                                       |
|-----------------------------|-----------------------|---------------------------------------------------------------|
| Total interactions          | 5,002                 | 4-year window, excl. departed & bots                          |
| Unique contributors         | 210                   | 4-year window; 46 (21.9%) appear only once                    |
| Bus factor (4-year)         | 2                     | baentsch + RodriM11 = 50% of 4-year activity                  |
| Bus factor (6-month)        | 2                     | RodriM11 37.2% + baentsch 25.4% — both now active Maintainers |
| Top contributor (4yr)       | baentsch<br>47.0%     | Critical concentration; 25.4% in 6-month window — improving   |
| Top contributor (6mo)       | RodriM11<br>37.2%     | Major structural shift: RodriM11 now primary driver           |
| Top-3 share (4yr)           | 59.2%                 | High concentration; RodriM11 now in top 3                     |
| One-time contributors (4yr) | 21.9% (46)            | Similar to liboqs                                             |
| Retained ≥3 months (4yr)    | 25.2%                 | Stable                                                        |
| Peak month                  | Jan 2025 (248 events) | Post-LoA baentsch return peak                                 |
| Event trend (YoY 2024→2025) | −46.1%                | Severe — stabilising in late 2025/early 2026                  |
| Active streak ≥10 contrib.  | 16 months             | Ended Jan 2026; since then 5–14/month                         |
| New contributors 2024       | 83                    | Strong intake year                                            |
| New contributors 2025       | 43                    | Continued decline; 2026 on pace for ≈32                       |

#### 3.2 The Transition: From Single-Maintainer to Dual-Maintainer



The most significant development since March is the demonstrated parity of activity between RodriM11 and baentsch. Since September 2025 the two Maintainers have operated in near-parity for extended periods, with RodriM11 now consistently leading:

| Month    | baentsch events | RodriM11 events | Context                              |
|----------|-----------------|-----------------|--------------------------------------|
| 2025-09  | 75              | 18              | Both return; baentsch leads          |
| 2025-10  | 69              | 45              | Converging                           |
| 2025-11  | 30              | 20              | Parallel activity                    |
| 2025-12  | 27              | 20              | Near parity                          |
| 2026-01  | 26              | 19              | Near parity                          |
| 2026-02  | 56              | 54              | Near-identical — highest joint month |
| 2026-03  | 13              | 14              | RodriM11 leads for first time        |
| 2026-04  | 13              | 68              | RodriM11 dominant                    |
| 2026-05  | 14              | 14              | Parallel low                         |
| 2026-06* | 9               | 26              | RodriM11 leads (partial month)       |

\* June 2026 is a partial month at time of data export.

### 3.3 Monthly Activity — Last 18 Months

| Month   | Contributors | Events | Activity bar |
|---------|--------------|--------|--------------|
| 2024-12 | 26           | 266    | <div></div>  |
| 2025-01 | 27           | 248    | <div></div>  |
| 2025-02 | 20           | 135    | <div></div>  |
| 2025-03 | 14           | 67     | <div></div>  |
| 2025-04 | 14           | 61     | <div></div>  |
| 2025-05 | 13           | 75     | <div></div>  |
| 2025-06 | 10           | 30     | <div></div>  |

| Month    | Contributors | Events | Activity bar |
|----------|--------------|--------|--------------|
| 2025-07  | 9            | 34     | <div></div>  |
| 2025-08  | 11           | 33     | <div></div>  |
| 2025-09  | 14           | 122    | <div></div>  |
| 2025-10  | 9            | 131    | <div></div>  |
| 2025-11  | 13           | 88     | <div></div>  |
| 2025-12  | 11           | 75     | <div></div>  |
| 2026-01  | 12           | 75     | <div></div>  |
| 2026-02  | 8            | 136    | <div></div>  |
| 2026-03  | 9            | 44     | <div></div>  |
| 2026-04  | 8            | 141    | <div></div>  |
| 2026-05  | 5            | 55     | <div></div>  |
| 2026-06* | 7            | 68     | <div></div>  |

Events include all engagement types including bot-authored commit comments. Contributors show unique human actors including departed. \* June 2026 is a partial month.

3.4 Key Contributors (6-Month Window: Dec 2025 – Jun 2026)

| Contributor | Events | Share | Active Months | Role / Notes                                        |
|-------------|--------|-------|---------------|-----------------------------------------------------|
| RodriM11    | 195    | 37.2% | 6/7           | Maintainer (elevated Mar 2026) — now primary driver |
| baentsch    | 133    | 25.4% | 7/7           | Maintainer — declining trend; still essential       |
| ashman-p    | 72     | 13.7% | 6/7           | Maintainer — see review standards discussion (G10)  |
| Vishnu2707  | 29     | 5.5%  | 3/7           | Rising newcomer; cross-repo                         |
| dtebila     | 11     | 2.1%  | 1/7           | Release Manager; primarily liboqs                   |
| rben-dev    | 10     | 1.9%  | 1/7           | Occasional cross-repo                               |

| Contributor | Events | Share | Active Months | Role / Notes |
|-------------|--------|-------|---------------|--------------|
| iiviel      | 9      | 1.7%  | 1/7           | Newcomer     |
| petrov      | 8      | 1.5%  | 1/7           | Occasional   |
| bob-beck    | 7      | 1.3%  | 2/7           | Occasional   |
| iyamv       | 7      | 1.3%  | 1/7           | Newcomer     |

## 4. Cross-Repository Ecosystem Analysis

### 4.1 Contributor Overlap

| Overlap Threshold                     | Count | Notes                                     |
|---------------------------------------|-------|-------------------------------------------|
| Any account in both repos (raw, 4yr)  | 77    | Includes single-event cross-touches       |
| Meaningful: ≥3 events in each repo    | 46    | Casual cross-participation                |
| Genuine: ≥10 events in each repo      | 21    | True dual-repo contributors               |
| liboqs only (4yr, any activity)       | 257   | Large untapped pool for provider bridging |
| oqs-provider only (4yr, any activity) | 133   | Provider specialists                      |

abhi-dev-engg (116 liboqs events in 6 months, 0 in oqs-provider) and xuganyu96 (184 liboqs events, 0 in provider) remain the two highest-priority bridge targets. Both have demonstrated technical depth in liboqs and no cross-repo presence.

### 4.2 Review Culture

| Repository   | Approved  | Changes Requested | Comment-only | Total Reviews (4yr) |
|--------------|-----------|-------------------|--------------|---------------------|
| liboqs       | 730 (32%) | 77 (3%)           | 1,404 (62%)  | 2,259               |
| oqs-provider | 297 (33%) | 44 (5%)           | 550 (61%)    | 904                 |

The low changes-requested rate (3–5%) is worth monitoring. It can reflect a healthy pre-screening culture (PRs discussed before submission) or reviewer passivity. The inline comment body-size data in section 4.2b provides the clarifying evidence.

#### 4.2a Inline/Approval Ratio by Year

The ratio of inline PR review comments to approval verdicts is a proxy for review rigour. A falling ratio indicates reviewers approving without substantive engagement.

| Year | Approvals | Inline Comments | Ratio | Assessment                                      |
|------|-----------|-----------------|-------|-------------------------------------------------|
| 2019 | 50        | 40              | 0.80x | Low                                             |
| 2020 | 147       | 69              | 0.47x | Lowest — bandwidth pressure during rapid growth |
| 2021 | 187       | 202             | 1.08x | Recovered                                       |

| Year  | Approvals | Inline Comments | Ratio | Assessment                               |
|-------|-----------|-----------------|-------|------------------------------------------|
| 2022  | 129       | 176             | 1.36x | Good                                     |
| 2023  | 284       | 533             | 1.88x | Strong                                   |
| 2024  | 440       | 1,193           | 2.71x | Best year — most rigorous                |
| 2025  | 355       | 547             | 1.54x | Healthy                                  |
| 2026* | 125       | 348             | 2.78x | Very high (partial year — strong signal) |

4.2b Reviewer Profile Analysis (4-Year Window)

Body text measured after stripping quoted lines, code blocks, images, and HTML comments. Only original new prose is counted.

| Reviewer  | Approvals | Empty% | Appr Med | ChgReq% | Inline Med | IC Med | Assessment                                     |
|-----------|-----------|--------|----------|---------|------------|--------|------------------------------------------------|
| baentsch  | 315       | 38%    | 35c      | 6.6%    | 125c       | 262c   | Best combined — writes summaries               |
| dtebila   | 425       | 98%    | 0c       | 2.5%    | 72c        | 151c   | High empty%; inline substance offsets          |
| bhess     | 116       | 56%    | 0c       | 0.7%    | 102c       | 196c   | Acceptable — healthy inline size               |
| ashman-p  | 70        | 99%    | 0c       | 0.0%    | 27c        | 155c   | Lowest on all three metrics — see G10          |
| xuganyu96 | 29        | 59%    | 0c       | 18.3%   | 172c       | 241c   | Best chg_req% — most rigorous reviewer         |
| RodriM11  | 8         | 100%   | 0c       | 2.4%    | 143c       | 230c   | Strong inline; low approval count (early role) |

The reviewer-level variance shown above — from baentsch’s combined profile (38% empty, 125c inline) to the lowest end of the range (99% empty, 27c inline, no changes-requested across 72 approvals) — illustrates why a shared definition of review quality matters: without one, two reviewers can both be ‘meeting expectations’ while operating on entirely different standards. This is a structural gap to close, not an individual performance issue.

## 5. Recommendations

---

Recommendations are grounded in the June 2026 data and expanded to five per urgency tier, ordered within each tier by expected impact on the health scorecard (section 7). Each item names the dimension(s) it targets and the metric that would move if implemented — this is the basis for the ordering, not a subjective ranking. Items fulfilled since March are noted where relevant.

### 5.1 Critical — Address Immediately

#### P1 — Document baentsch’s LoA end date [G5]

**Targets:** Succession readiness, Bus factor / resilience. baentsch declared a leave of absence in liboqs on March 11, 2025. The governance document caps leaves at one year (maximum: March 2026). The LoA is now 15 months old and technically non-compliant with the project’s own rules, while baentsch remains the most active combined contributor with 269 events across both repos in the 6-month window. Resolution: formally end the LoA and restore voting rights, or initiate a documented extension. Closing this gap directly improves the succession readiness sub-score, currently the most-improved but still incomplete dimension.

#### P2 — Establish a shared definition of review quality [G10]

**Targets:** Review culture, Succession readiness. The data shows wide variance in what ‘approval’ means in practice across reviewers: empty-body approval rates range from 36% to 99%, median inline comment length from 27c to 172c, and changes-requested rates from 0% to 18.3% (see 4.2b). This variance is itself the finding — it suggests the project has never explicitly defined what a substantive review looks like, leaving each reviewer to set their own bar. A short Discussion thread proposing minimum expectations (e.g. a non-empty summary on approval, or a documented exception for trivial PRs) would give every Maintainer and Committer a shared standard to hold themselves to. This is the natural next step now that oqs-provider has the second Maintainer (RodriM11) needed to make the conversation collaborative rather than top-down. Raises the review culture sub-score in both repos and reduces single-reviewer risk ahead of any future maintainer transition.

#### P3 — Initiate Committer vote for abhi-dev-engg in liboqs [G7 partial]

**Targets:** Succession readiness, Bus factor / resilience. abhi-dev-engg has 116 events in the last 6 months — more than bhes (106), mkannwischer (26), and rtjk (25) — with no formal role in either repo. This is the clearest role-vs-activity gap in the dataset. The GOVERNANCE.md criteria (‘sufficient code and deep subject matter knowledge’) are demonstrably met. Formalising this role widens the liboqs bench by one and closes the single largest activity/governance mismatch currently outstanding.

#### P4 — Write Architecture Decision Records for oqs-provider [G9]

**Targets:** Succession readiness, Bus factor / resilience. Institutional knowledge is now the primary risk in both repos. baentsch holds 58 months of oqs-provider design history that exists nowhere in writing. Even three or four ADRs covering the highest-impact design decisions would materially reduce this risk. A 1–2 day investment with multi-year payoff, and the single most direct mitigation against the irreplaceable-knowledge risk identified in Scenario A and B (section 6).

#### P5 — Reduce oqs-provider’s top-1 concentration below 35% within two quarters

**Targets:** Bus factor / resilience, Contributor breadth. baentsch and RodriM11 together hold 63.3% of 6-month oqs-provider activity — healthier than March’s single-maintainer 47–48%, but still concentrated in two people. A concrete numeric target (e.g. no single contributor above 35% of monthly events for two consecutive quarters) gives the existing dual-Maintainer structure something measurable to work toward, and would be the clearest signal that P1–P4 have actually changed the risk profile rather than just the org chart.

## 5.2 High Priority — Address Within 3 Months

### P6 — Bridge xuganyu96 and abhi-dev-engg into oqs-provider

**Targets:** Cross-repo connectivity, Bus factor / resilience. xuganyu96 (184 liboqs events in 6 months, 18.3% changes-requested rate, 172c inline median) has zero oqs-provider events. abhi-dev-engg (116 liboqs events) similarly. A targeted Discussion post in oqs-provider with labelled issues suited to their liboqs expertise is the recommended mechanism. Directly grows the pool of  $\geq 10$ -event dual-repo contributors (currently 21) and gives oqs-provider two more candidate reviewers with proven rigour.

### P7 — Implement structured first-contribution pathway

**Targets:** Retention rate, New contributor intake. The 2024–2025 cohorts retain at 19–21%, below the 30–36% of 2019–2021. Specific mechanism: labelled ‘good first issue’ tickets in oqs-provider, a provider-specific contribution guide, and a personal follow-up comment on every first PR within 7 days. Active contributors crossing the 3-month threshold have historically shown 4× higher long-term retention — this is the highest-leverage lever on the lowest-scoring dimension in the scorecard (5.5/10 in both repos).

### P8 — Publish a 12-month post-NIST roadmap

**Targets:** New contributor intake, Growth trend. 103 new liboqs contributors in 2024 and 83 in 2025 signal continued external interest, but contributors arriving after NIST standardisation have no public map of where help is needed. A roadmap in GitHub Discussions tied to open issues and labelled by contributor tier would convert ambient interest into directed contribution, directly addressing the post-NIST growth slowdown (–28.3% liboqs, –46.1% provider YoY).

### P9 — Apply the 90-day inactivity rule consistently and proactively

**Targets:** Succession readiness. The March report found three Committers/Maintainers past the 90-day inactivity threshold without the rule being applied; this quarter’s correction (G1–G4) shows the mechanism works once triggered, but currently relies on someone noticing. Automating a quarterly check against the governance file (cross-referencing last-activity dates) would prevent the same staleness from recurring and keep GOVERNANCE.md continuously trustworthy rather than periodically corrected.

### P10 — Track and publish the inline/approval ratio quarterly

**Targets:** Review culture. The inline/approval ratio has been the single most informative review-rigour metric in this report (0.47x in 2020 to 2.78x in 2026), yet it is not currently tracked by the project itself. Publishing it alongside the review-quality standard proposed in P2 would let the project see drift before it becomes a 4-year pattern, and gives the standard from P2 a number to check itself against.

## 5.3 Medium Term — Address Within 6 Months

### P11 — Monthly automated health pulse

**Targets:** Bus factor / resilience, Succession readiness. A one-page automated monthly summary — active contributor count, new contributors, review throughput, median time-to-first-response — costs one hour to implement and creates lasting accountability. The departure patterns of SWilson4 and praveksharma would have been visible months earlier with this mechanism, and it is the cheapest available early-warning system for the concentration risk this report tracks every quarter.

### P12 — Formalise a written succession plan in GOVERNANCE.md

**Targets:** Succession readiness. Beyond the individual role changes already made (P1, P3), neither repository names backup owners for specific subsystems or documents who would step in if a given Maintainer became unavailable without notice. A short table in GOVERNANCE.md — area, current owner, named backup — converts the ad hoc successions already happening in practice (RodriM11, xuganyu96) into a documented, repeatable process.

---

### P13 — Introduce a lightweight mentorship pairing for new Core-tier contributors

**Targets:** Retention rate, New contributor intake. Vishnu2707 (oqs-provider, 5.5% of 6-month activity, 3 active months) and rben-dev (liboqs, 2.3%, 4 active months) are the clearest examples of contributors crossing from Occasional into Regular activity right now. Pairing rising contributors with a Core-tier reviewer for their first 2–3 months — explicit PR assignment, faster review turnaround — would test whether the onboarding friction identified in P7 is structural or simply a matter of attention.

### P14 — Revisit the bot-filtering convention in CI-generated commit comments

**Targets:** Review culture (data quality). github-actions[bot] alone accounts for the majority of raw commit-comment volume in both repos, which inflates total event counts without reflecting human engagement (see section 1 methodology note). Tightening CI to comment only on failures, or moving routine status into checks rather than comments, would make the project's own activity signals cleaner — a small technical change that improves the reliability of every future health report.

### P15 — Establish a standing quarterly governance review

**Targets:** Succession readiness, Bus factor / resilience. This report and its March predecessor have each surfaced governance drift that had accumulated silently for months. Formalising this report's cadence — a standing agenda item at a quarterly OQS TSC meeting to review GOVERNANCE.md against the latest activity data — turns a reactive correction exercise into a routine one, and is the structural reason the next report should show fewer items moving from 'Open' to 'Critical' between cycles.



## 6. What-If Analysis — Key Maintainer Departure Scenarios

Based on the 6-month trailing window (Dec 25, 2025 – Jun 25, 2026), bots and departed excluded. The effective active community is 16 people: 9 Core (≥30 events, ≥3 months) and 7 Regular (≥10 events, ≥2 months).

### 6.0 Current Active Community (Combined, 6-Month Window)

| Tier       | Definition            | Count | Who                                                                                                                                               |
|------------|-----------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Core       | ≥30 events, ≥3 months | 9     | baentsch (269), dstebila (261), RodriM11 (226), xuganyu96 (184), ashman-p (141), abhi-dev-engg (116), bhess (108), Vishnu2707 (45), rben-dev (38) |
| Regular    | ≥10 events, ≥2 months | 7     | mkannwischer (26), rtjk (25), bbbrumley (24), pablo-gf (19), mdeilman (13), Frauschi (12), embetrix (11)                                          |
| Occasional | ≥5 events             | ~12   | Recurring but low-volume participants                                                                                                             |
| Peripheral | 1–4 events            | ~31   | One-time commenters, single PRs — not recurring                                                                                                   |

Combined 6-month bus factor: 4. Top 4 (baentsch 15.1%, dstebila 14.6%, RodriM11 12.7%, xuganyu96 10.3%) account for 52.7% of combined activity.

#### 6.1 Scenario A: baentsch Reduces Availability Further

baentsch’s monthly activity has declined from the Sep 2025 post-LoA peak (110 provider events) to 9–14 events/month in Apr–Jun 2026. Assessment: substantially less critical than March. RodriM11’s Maintainer elevation means oqs-provider has a documented successor with demonstrated throughput. The primary remaining gap is institutional knowledge (G9), not merge capacity.

| Impact Area             | Estimated Loss       | Absorption Capacity                                           |
|-------------------------|----------------------|---------------------------------------------------------------|
| oqs-provider events     | ~133/6mo (25.4%)     | RodriM11 (37.2%) and ashman-p (13.7%) cover merge authority   |
| liboqs events           | ~136/6mo (11.4%)     | dstebila, xuganyu96, abhi-dev-engg absorb; minimal disruption |
| Approval verdicts       | ~30% of oqs-provider | RodriM11 has merge authority; bhess and ashman-p supplement   |
| Institutional knowledge | Significant          | ADRs not yet written (G9 open) — primary remaining risk       |

#### 6.2 Scenario B: dstebila Reduces Availability

dstebila now leads the 6-month liboqs window at 20.9% (250 events) — a larger share than at the time of the March analysis. Departure would be more impactful than March’s scenario B suggested.

| Impact Area   | Estimated Loss            | Absorption Capacity                                                       |
|---------------|---------------------------|---------------------------------------------------------------------------|
| liboqs events | 250 events (20.9% of 6mo) | xuganyu96 (184), baentsch (136), abhi-dev-engg (116) provide strong bench |

| Impact Area          | Estimated Loss           | Absorption Capacity                                                     |
|----------------------|--------------------------|-------------------------------------------------------------------------|
| oqs-provider events  | 11 events (2.1%)         | Negligible                                                              |
| Approval verdicts    | ~47% of liboqs 4yr total | xuganyu96 (18.3% chg_req rate) and abhi-dev-engg are natural successors |
| Co-founder knowledge | Irreplaceable            | 119 months of architectural history; ADRs would substantially mitigate  |

## 7. Community Health Scorecard

Scores reflect community engagement health only — not code quality, security, or technical completeness. Primarily based on the 6-month trailing window; 4-year context noted where relevant.

| Dimension               | liboqs | oqs-provider | Weight | Notes                                                                           |
|-------------------------|--------|--------------|--------|---------------------------------------------------------------------------------|
| Contributor breadth     | 7.5    | 4.0          | High   | liboqs 13–33/month; provider 8–27/month (Dec 2024 spike)                        |
| Bus factor / resilience | 6.5    | 4.0          | High   | liboqs BF=4 on 6mo (improved); provider BF=2 but dual-Maintainer now real       |
| Growth trend            | 6.5    | 4.0          | Medium | liboqs –28.3% YoY; provider –46.1% YoY                                          |
| New contributor intake  | 7.5    | 5.5          | Medium | liboqs 83 new in 2025; provider 43 new in 2025                                  |
| Retention rate          | 5.5    | 5.5          | High   | 2024–2025 cohorts at 19–21%; below 2019–2021 benchmark                          |
| Review culture          | 7.0    | 5.5          | Medium | Inline ratio strong (2.78x in 2026); ashman-p concern persists                  |
| Cross-repo connectivity | 6.0    | 5.0          | Low    | 21 genuine dual-active; RodriM11 bridging at high volume                        |
| Succession readiness    | 6.0    | 6.5          | High   | xuganyu96 + RodriM11 now formally recognised; provider has 3 active Maintainers |
| COMPOSITE SCORE         | 6.5    | 5.1          | —      | 4-year window, excl. departed; all event types per March methodology            |

## 8. Governance Alignment — GOVERNANCE.md vs Activity Reality

Both GOVERNANCE.md documents were read live from the repositories on June 25, 2026 and cross-referenced against activity data. The March 2026 report identified nine governance items (G1–G9); a tenth (G10) is introduced this quarter. Status of each is tracked below.

| Item                                      | March Status | June Status           | Evidence                                                               |
|-------------------------------------------|--------------|-----------------------|------------------------------------------------------------------------|
| G1: liboqs<br>SWilson4 removal            | Required     | ✓ Resolved            | SWilson4 → Emeritus Maintainer in GOVERNANCE.md                        |
| G2: liboqs<br>Martyrshot 90-day           | Required     | ✓ Resolved            | Martyrshot → Emeritus Committer                                        |
| G3: liboqs<br>praveksharma 90-day         | Required     | ✓ Resolved            | praveksharma → Emeritus Committer                                      |
| G4:<br>christianpaquin /<br>vssoftco      | Recommended  | Resolved              | Both listed as Emeritus Committers                                     |
| G5: baentsch LoA<br>clarification         | Recommended  | ⚠ Worsened            | LoA now 15 months — exceeds 12-month governance cap                    |
| G6: RodriM11<br>Committer liboqs          | Recommended  | ✓ Resolved            | RodriM11 listed as Committer in liboqs GOVERNANCE.md                   |
| G7: xuganyu96 /<br>abhi-dev-engg          | Recommended  | Partially<br>resolved | xuganyu96 → Maintainer liboqs; abhi-dev-engg still without formal role |
| G8: RodriM11<br>Maintainer<br>provider    | Critical     | ✓ Resolved            | RodriM11 listed as Maintainer in oqs-provider — highest-impact change  |
| G9: Succession<br>documentation           | Recommended  | Open                  | No ADRs or succession plan written in either repo                      |
| G10: shared<br>review-quality<br>standard | —            | NEW                   | No documented minimum review expectation across either repo            |

### 8.1 Current Named Role-Holders

| Person        | liboqs Role                  | oqs-provider Role     | 6-mo Activity | Status                                         |
|---------------|------------------------------|-----------------------|---------------|------------------------------------------------|
| baentsch      | Maintainer, Committer (LoA?) | Maintainer, Committer | 136+133 = 269 | Active — LoA exceeds maximum (G5)              |
| dtebila       | Maintainer, Committer        | Release Manager       | 250+11 = 261  | Active — leading liboqs contributor            |
| xuganyu96     | Maintainer, Committer        | Release Manager       | 184+0 = 184   | Active — newly formalised; bridge opportunity  |
| ashman-p      | Committer                    | Maintainer, Committer | 69+72 = 141   | Active — see review standards discussion (G10) |
| bhess         | Committer                    | Committer             | 106+2 = 108   | Active — solid                                 |
| mkannwischer  | Committer                    | —                     | 26+0 = 26     | Active                                         |
| RodriM11      | Committer                    | Maintainer, Committer | 31+195 = 226  | Active — primary oqs-provider driver           |
| abhi-dev-engg | —                            | —                     | 116+0 = 116   | Active — no formal role (G7 open)              |

## 9. Quarter-on-Quarter Comparison: March vs June 2026

Direct comparison of March 14, 2026 figures against June 25, 2026 figures. ↑ improvement, ↓ decline, → unchanged. Both reports use the same event-counting methodology: all event types including bot-authored commit comments for volume; human actors (including departed) for monthly contributor counts; human actors excluding departed for all health metrics.

### 9.1 Metrics Comparison

| Metric                               | March 14, 2026   | June 25, 2026    | Δ           | Notes                                                                                          |
|--------------------------------------|------------------|------------------|-------------|------------------------------------------------------------------------------------------------|
| [liboqs] All-time events             | 17,472           | 18,214           | ↑<br>+742   | 3.5 months of new events; same methodology                                                     |
| [liboqs] All-time contributors       | 426              | 437              | ↑ +11       | Net new contributors since March                                                               |
| [liboqs] 4yr total events            | 9,011            | 9,189            | ↑<br>+178   | Window extended; discussions + pr_merged now included for both reports                         |
| [liboqs] 4yr contributors            | 330              | 334              | ↑ +4        | Continued intake                                                                               |
| [liboqs] Bus factor (4yr)            | 3                | 3                | →           | Stable                                                                                         |
| [liboqs] Bus factor (6mo)            | N/A              | 4                | NEW<br>↑    | 6mo window introduced this quarter                                                             |
| [liboqs] Top contributor share (4yr) | 28.2% (baentsch) | 27.4% (baentsch) | ↑<br>+0.8pp | Slowly reducing                                                                                |
| [liboqs] Top contributor (6mo)       | baentsch         | dstebila 20.9%   | ↑<br>Major  | Leadership rotation confirmed                                                                  |
| [liboqs] Active streak ≥10           | 40 months        | 43 months        | ↑<br>+3mo   | Streak extended and still ongoing                                                              |
| [liboqs] YoY event change 2024→2025  | reported: -15.8% | -28.3%           | ↓           | Different base: March used the 4yr total excl. discussions; now consistent with full event set |
| [liboqs] New contributors 2024       | 104              | 103              | →           | Stable                                                                                         |
| [liboqs] New contributors 2025       | 84               | 83               | →           | Stable                                                                                         |

| Metric                           | March 14, 2026   | June 25, 2026    | Δ        | Notes                                                    |
|----------------------------------|------------------|------------------|----------|----------------------------------------------------------|
| [liboqs] Retention ≥3mo 2024     | 22%              | 19%              | ↓        | Full-year 2025 data shows fewer returns from 2024 cohort |
| [liboqs] RodriM11 role           | Not listed       | Committer        | ↑ G6     | Resolved                                                 |
| [liboqs] xuganyu96 role          | None             | Maintainer       | ↑ G7     | Resolved                                                 |
| [provider] All-time events       | 5,310            | 5,658            | ↑ +348   | Same methodology; 3.5 months new events                  |
| [provider] All-time contributors | 210              | 217              | ↑ +7     | Net new contributors                                     |
| [provider] 4yr total events      | 4,704            | 5,002            | ↑ +298   | Window extended + discussions added                      |
| [provider] Bus factor (4yr)      | 2                | 2                | →        | Nominally unchanged; structural risk reduced             |
| [provider] Top contributor (4yr) | 48.5% (baentsch) | 47.0% (baentsch) | ↑ +1.5pp | Improving slowly                                         |
| [provider] Top contributor (6mo) | baentsch ~48%    | RodriM11 37.2%   | ↑ Major  | RodriM11 now leading                                     |
| [provider] RodriM11 role         | Committer        | Maintainer       | ↑ G8     | Most impactful change since project inception            |
| [provider] YoY 2024→2025         | -54.0%           | -46.1%           | ↑ +7.9pp | Less severe than projected                               |
| [Gov] G1 SWilson4                | Required         | ✓ Resolved       | ↑        |                                                          |
| [Gov] G2 Martyrshot              | Required         | ✓ Resolved       | ↑        |                                                          |
| [Gov] G3 praveksharma            | Required         | ✓ Resolved       | ↑        |                                                          |
| [Gov] G8 RodriM11 Maintainer     | Critical         | ✓ Resolved       | ↑        | Highest-impact item from March                           |
| [Gov] G5 baentsch LoA            | Recommended      | ⚠ Non-compliant  | ↓        | Now exceeds 12-month cap                                 |

9.2 Score Delta Summary

| Dimension               | liboqs Mar | liboqs Jun | Δ    | Provider Mar | Provider Jun | Δ    |
|-------------------------|------------|------------|------|--------------|--------------|------|
| Contributor breadth     | 7.5        | 7.5        | →    | 4.5          | 4.0          | −0.5 |
| Bus factor / resilience | 6.0        | 6.5        | +0.5 | 3.0          | 4.0          | +1.0 |
| Growth trend            | 7.0        | 6.5        | −0.5 | 4.0          | 4.0          | →    |
| New contributor intake  | 7.5        | 7.5        | →    | 6.0          | 5.5          | −0.5 |
| Retention rate          | 6.0        | 5.5        | −0.5 | 5.5          | 5.5          | →    |
| Review culture          | 7.0        | 7.0        | →    | 5.5          | 5.5          | →    |
| Cross-repo connectivity | 6.0        | 6.0        | →    | 4.5          | 5.0          | +0.5 |
| Succession readiness    | 5.0        | 6.0        | +1.0 | 2.5          | 6.5          | +4.0 |
| COMPOSITE               | 6.7        | 6.5        | −0.2 | 4.3          | 5.1          | +0.8 |

The liboqs composite decline of 0.2 reflects post-NIST normalisation and the full-year retention data showing 2024–2025 cohorts performing below earlier benchmarks. The oqs-provider improvement of 0.8 is structural: five governance items resolved in one quarter, dual-Maintainer now operational. The succession readiness jump (+4.0) is the largest single-quarter improvement in any dimension across either repository.



---

## 10. Closing Assessment

---

The OQS ecosystem has made its most significant governance improvements since the project’s founding in the three months since the March 2026 report. The elevation of RodriM11 to Maintainer in oqs-provider, xuganyu96 to Maintainer in liboqs, and the formalisation of RodriM11 as Committer in liboqs represent a genuine second tier of leadership where none existed before. These changes are not reflected only in the governance documents — they are visible in the event data: RodriM11 now carries the highest event share in oqs-provider, and xuganyu96 drives liboqs as its second most active contributor.

liboqs shows a stronger structural picture than the March report’s headline numbers suggested. The active streak has extended to 43 months (still ongoing), and the May 2026 activity spike (406 events) demonstrates continued external engagement. The YoY decline of  $-28.3\%$  is real but expected given the exceptional 2024 NIST peak; what matters is that breadth — the number of distinct contributors per month — has remained healthy at 13–33.

The concerns that remain are specific and actionable. The absence of a shared review-quality standard (G10) is a clear gap worth closing: the data shows a wide and unaddressed spread in what ‘approval’ means across reviewers. The baentsch LoA documentation issue has crossed from recommended clarification to technical non-compliance. The ADR gap (G9) is the one item that will look obvious in hindsight if a key maintainer departs without it. Of the fifteen recommendations in section 5, the five Critical items (P1–P5) target the dimensions with the most room to move — succession readiness and bus factor / resilience — and are achievable within weeks rather than months. Addressing them would bring both repositories to a governance standard commensurate with their technical maturity and the critical infrastructure they support.

---

*Data covers community engagement events (issue comments, PR review inline comments, PR review verdicts, PR opened, PR merged, discussion threads and replies, commit comments) — raw commits, code review diffs, and CI results are out of scope. All figures derived from GitHub REST API v3 live data pull, June 25, 2026. Event totals include bot-authored commit comments for volume consistency with the March 2026 report; contributor counts and health metrics exclude bot actors.*